Exercise 2 Date:

Develop and compare CLI, GUI, and Voice User Interfaces (VUI) for the same task and assess user satisfaction using Python (Tkinter for GUI, Speech Recognition for VUI), Terminal

AIM:

The aim is to develop and compare Command Line Interface (CLI), Graphical User Interface (GUI), and Voice User Interface (VUI) for the same task, and assess user satisfaction using Python (with Tkinter for GUI and Speech Recognition for VUI) and Terminal.

PROCEDURE:

i) CLI (Command Line Interface)

CLI implementation where users can add, view, and remove students.

```
 students = []

def add_student(name):

    students.append(name)

    print(f"Student '{name}' added successfully.")

def view_students():

    if students:

        print("\n📋 Student List:")

        for idx, student in enumerate(students, 1):
```

```
            print(f"{idx}. {student}")
    else:
        print("No students found.")
def remove_student(number):
    if 0 < number <= len(students):
        removed = students.pop(number - 1)
        print(f"Student '{removed}' removed successfully.")
    else:
        print("Invalid number.")
def main():
    while True:
        print("\n===== Student Management System =====")
        print("1. Add Student")
        print("2. View Students")
        print("3. Remove Student")
        print("4. Exit")

        choice = input("Enter your choice (1-4): ")
        if choice == '1':
            name = input("Enter student name: ")
```

```
            add_student(name)
        elif choice == '2':
            view_students()
        elif choice == '3':
            try:
                num = int(input("Enter student number to remove: "))
                remove_student(num)
            except ValueError:
                print("Please enter a valid number.")
        elif choice == '4':
            print("Exiting program...")
            break
        else:
            print("Invalid choice. Please try again.")


if __name__ == "__main__":
    main()
```

Sample Output:

```
PS C:\Users\vkavy\OneDrive\Attachments\Desktop\cli.py> python gui.py
PS C:\Users\vkavy\OneDrive\Attachments\Desktop\cli.py> python todo.py

Options:
1. Add Student
2. View Students
3. Remove Student
4. Exit
Enter your choice: 1
Enter student name: joe
Student 'joe' added successfully.

Options:
1. Add Student
2. View Students
3. Remove Student
4. Exit
Enter your choice: 2
Student List:
1. joe

Options:
1. Add Student
2. View Students
3. Remove Student
4. Exit
Enter your choice: 3
Enter student number to remove: joe
```

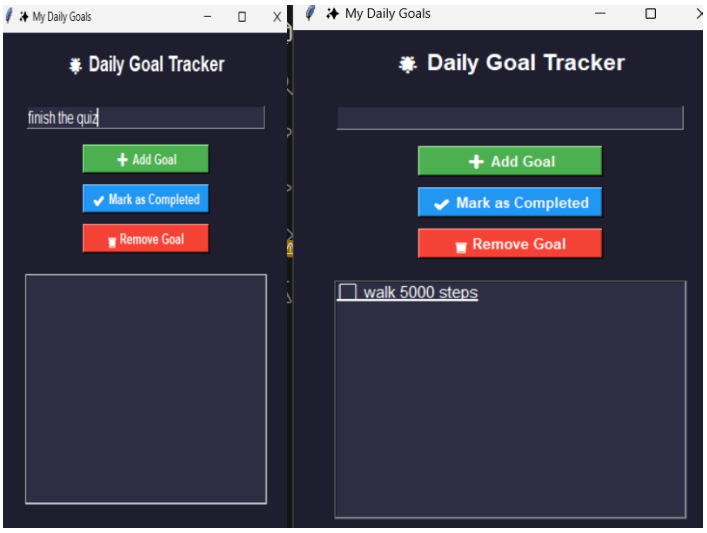
ii) GUI (Graphical User Interface)

Tkinter to create a simple GUI for our To-Do List application.

```
import tkinter as tk

from tkinter import messagebox

goals = []

def add_goal():

    text = entry.get()

    if text:

        goals.append(text)

        listbox.insert(tk.END, text)

        entry.delete(0, tk.END)

    else:
```

```
        messagebox.showwarning("Warning", "Enter a goal!")

def remove_goal():

    selected = listbox.curselection()

    if selected:

        listbox.delete(selected)

        goals.pop(selected[0])

root = tk.Tk()

root.title("Simple Goal Tracker")

root.geometry("300x350")

entry = tk.Entry(root, width=30)

entry.pack(pady=10)

tk.Button(root, text="Add Goal", command=add_goal).pack(pady=5)

tk.Button(root, text="Remove Goal",
command=remove_goal).pack(pady=5)



listbox = tk.Listbox(root, width=35, height=10)

listbox.pack(pady=10)

root.mainloop()
```

Sample Output:



iii) VUI (Voice User Interface)

speech_recognition library for voice input and the pyttsx3 library for text-to-speech output. Make sure you have these libraries installed (pip install SpeechRecognition pyttsx3).

```
import speech_recognition as sr
import pyttsx3

tasks = []
recognizer = sr.Recognizer()
engine = pyttsx3.init()

def add_task(task):
    tasks.append(task)
    engine.say(f"Task {task} added")
    engine.runAndWait()

def view_tasks():
    if tasks:
        engine.say("Your tasks are")
        for task in tasks:
            engine.say(task)
    else:
        engine.say("No tasks to show")
    engine.runAndWait()

def remove_task(task_number):
```

```
    if 0 < task_number <= len(tasks):
        removed_task = tasks.pop(task_number - 1)
        engine.say(f"Task {removed_task} removed")
    else:
        engine.say("Invalid task number")
    engine.runAndWait()

def recognize_speech():
    with sr.Microphone() as source:
        print("Listening...")
        audio = recognizer.listen(source)
        try:
               command = recognizer.recognize_google(audio)
            return command
        except sr.UnknownValueError:
            engine.say("Sorry, I did not understand that")
            engine.runAndWait()
            return None

def main():
    while True:
        engine.say("Options: add task, view tasks, remove
task, or exit")
        engine.runAndWait()

        command = recognize_speech()
        if not command:
            continue

        if "add task" in command:
            engine.say("What is the task?")
            engine.runAndWait()
            task = recognize_speech()
            if task:
                add_task(task)
        elif "view tasks" in command:
            view_tasks()
        elif "remove task" in command:
                 engine.say("Which task number to remove?")
            engine.runAndWait()
            task_number = recognize_speech()
```

```
            if task_number:
                    remove_task(int(task_number))
        elif "exit" in command:
            engine.say("Exiting...")
            engine.runAndWait()
            break
        else:
               engine.say("Invalid option. Please try again.")
            engine.runAndWait()

if name_== " main ":
    main()
```

Sample Output:

```
(Voice) Options: add task, view tasks, remove task, or exit
(User) add task
(Voice) What is the task?
(User) Finish report
(Voice) Task Finish report added
```

```
(Voice) Options: add task, view tasks, remove task, or exit
(User) view tasks
(Voice) Your tasks are: Finish report
```

```
(Voice) Options: add task, view tasks, remove task, or exit
(User) remove task
(Voice) Which task number to remove?
(User) 1
(Voice) Task Finish report removed
```